Transport–Spectral RH Program: Established Spine, Conditional Bridge, and Exact Missing Tasks

Abstract

This document consolidates the current rigorous spine of the transport–spectral contradiction program aimed at the Riemann Hypothesis. The established parts are: (i) Gaussian self-duality, theta symmetry, and the functional equation; (ii) the excess-transport formalism; (iii) vanishing excess under an exact window law; (iv) positive excess from an off-critical zero under a target bridge inequality. The resulting contradiction yields a clean conditional RH theorem. The remaining open part is isolated precisely as the transport-to-bank bridge.

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1 Status summary

The intended contradiction has the form

$$E_{\mathrm{ex}}(f) = 0 \qquad \text{and} \qquad \exists \rho \text{ off-critical} \Longrightarrow E_{\mathrm{ex}}(f) > 0.$$

If both statements are proved unconditionally for the same residue channel f, then the Riemann Hypothesis follows.

At the present stage, the clean situation is:

S1. The closure side has a usable standalone form: an exact window law implies $E_{\mathrm{ex}}(f) = 0$.

S2. The obstruction side has the correct coercive target: an off-critical zero should force a lower bound through an unbounded lifted drift penalty.

S3. The unresolved point is the rigorous bridge from the off-critical spectral component to the bank-level lower bound.

2 Gaussian self-duality and the functional equation

2.1 Gaussian self-duality

Lemma 2.1 (Gaussian self-duality)**.** *Let*

$$g(x) := e^{-\pi x^2}, \qquad \widehat{f}(\xi) := \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \xi}\, dx.$$

Then

$$\widehat{g}(\xi) = e^{-\pi \xi^2}.$$

Proof. Define

$$F(\xi) := \int_{-\infty}^{\infty} e^{-\pi x^2} e^{-2\pi i x \xi}\, dx.$$

Differentiate under the integral:

$$F'(\xi) = \int_{-\infty}^{\infty} e^{-\pi x^2} (-2\pi i x) e^{-2\pi i x \xi}\, dx.$$

Since

$$\frac{d}{dx} e^{-\pi x^2} = -2\pi x e^{-\pi x^2},$$

we may rewrite this as

$$F'(\xi) = i \int_{-\infty}^{\infty} \frac{d}{dx}(e^{-\pi x^2})\, e^{-2\pi i x \xi}\, dx.$$

Integrating by parts, the boundary term vanishes because $e^{-\pi x^2} \to 0$ as $|x| \to \infty$. Hence

$$F'(\xi) = -i \int_{-\infty}^{\infty} e^{-\pi x^2} \frac{d}{dx}(e^{-2\pi i x \xi})\, dx = -i \int_{-\infty}^{\infty} e^{-\pi x^2} (-2\pi i \xi) e^{-2\pi i x \xi}\, dx.$$

Thus

$$F'(\xi) = -2\pi \xi\, F(\xi).$$

Therefore F solves

$$F'(\xi) = -2\pi \xi F(\xi),$$

hence

$$F(\xi) = C e^{-\pi \xi^2}.$$

Evaluate at $\xi = 0$:

$$F(0) = \int_{-\infty}^{\infty} e^{-\pi x^2}\, dx = 1.$$

So $C = 1$, and therefore

$$\widehat{g}(\xi) = e^{-\pi \xi^2}.$$

□

2.2 Theta transformation

Lemma 2.2 (Theta transformation)**.** *Define*

$$\theta(t) := \sum_{n\in\mathbb{Z}} e^{-\pi n^2 t}, \qquad t > 0.$$

Then

$$\theta(t) = t^{-1/2}\theta(1/t).$$

Proof. Apply Poisson summation to

$$f_t(x) := e^{-\pi t x^2}.$$

By Gaussian self-duality with scaling,

$$\widehat{f}_t(\xi) = t^{-1/2} e^{-\pi\xi^2/t}.$$

Hence

$$\sum_{n\in\mathbb{Z}} e^{-\pi t n^2} = \sum_{m\in\mathbb{Z}} \widehat{f}_t(m) = t^{-1/2} \sum_{m\in\mathbb{Z}} e^{-\pi m^2/t}.$$

This is exactly

$$\theta(t) = t^{-1/2}\theta(1/t).$$

□

2.3 Functional equation

Theorem 2.3 (Functional equation of the completed zeta function)**.** *Define*

$$\xi(s) := \pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s).$$

Then ξ extends meromorphically and satisfies

$$\xi(s) = \xi(1-s).$$

Proof. For $\Re(s) > 1$,

$$\theta(t) - 1 = 2\sum_{n=1}^{\infty} e^{-\pi n^2 t}.$$

Multiply by $t^{s/2-1}$ and integrate on $(0,\infty)$:

$$\int_0^\infty (\theta(t)-1)t^{s/2-1}\,dt = 2\sum_{n=1}^{\infty}\int_0^\infty e^{-\pi n^2 t} t^{s/2-1}\,dt.$$

Substitute $u = \pi n^2 t$:

$$\int_0^\infty e^{-\pi n^2 t} t^{s/2-1}\,dt = (\pi n^2)^{-s/2}\Gamma\left(\frac{s}{2}\right).$$

Therefore

$$\int_0^\infty (\theta(t)-1)t^{s/2-1}\,dt = 2\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s),$$

hence

$$\xi(s) = \frac{1}{2}\int_0^\infty (\theta(t)-1)t^{s/2-1}\,dt.$$

Split at 1:

$$\xi(s) = \frac{1}{2}\int_0^1 (\theta(t)-1)t^{s/2-1}\,dt + \frac{1}{2}\int_1^\infty (\theta(t)-1)t^{s/2-1}\,dt.$$

On $(0,1)$, use

$$\theta(t) = t^{-1/2}\theta(1/t)$$

and substitute $t = 1/u$. After the standard rearrangement, the transformed integral becomes the same expression with s replaced by $1-s$. Hence

$$\xi(s) = \xi(1-s).$$

By analytic continuation, the identity extends meromorphically to all $s \in \mathbb{C}$. □

Corollary 2.4 (Symmetry of nontrivial zeros)**.** *If*

$$\xi(\rho) = 0,$$

then

$$\xi(1-\rho) = 0.$$

Hence the nontrivial zeros are symmetric with respect to the critical line

$$\Re(s) = \frac{1}{2}.$$

Proof. Immediate from the functional equation. □

3 Core objects of the excess framework

Definition 3.1 (Explicit residue distribution)**.** Let μ denote the explicit-formula residue distribution on $\mathbb{R}$ in a Guinand–Weil normalization.

Definition 3.2 (Dipole-regularized residue channel)**.** Fix a Schwartz atom $\psi \in \mathcal{S}(\mathbb{R})$ satisfying the dipole condition

$$\widehat{\psi}(0) = 0,$$

and define

$$f := \psi * \mu.$$

Definition 3.3 (Witness bank)**.** Fix dyadic scales

$$a_j = 2^j a_0, \qquad j = 0, \dots, J,$$

and positive weights $w_j > 0$. Define the second-difference witness response by

$$(W_a h)(\tau) := h(\tau + a) - 2h(\tau) + h(\tau - a).$$

Let $\varphi_{\tau,L}$ be a nonnegative Schwartz window localized on $[\tau, \tau + L]$. Define the bank energy

$$Q_{\text{bank}}(f; \tau, L) := \sum_{j=0}^{J} w_j^2 \, \langle W_{a_j} f, \varphi_{\tau,L} W_{a_j} f \rangle.$$

Definition 3.4 (Excess transport density)**.** Let $E_0 \geq 0$ be the baseline density. Define

$$E_{\text{ex}}(f) := \limsup_{L \to \infty} \sup_{\tau \in \mathbb{R}} \frac{Q_{\text{bank}}(f; \tau, L) - E_0 L}{L}.$$

4 Closure side

Definition 4.1 (Exact window law)**.** We say that f satisfies the exact window law if there exist

- a bounded function $J_f : \mathbb{R} \to \mathbb{R}$,
- a remainder $R(\tau, L)$,

such that

$$Q_{\text{bank}}(f; \tau, L) = E_0 L + J_f(\tau + L) - J_f(\tau) + R(\tau, L),$$

with

$$\sup_{\tau \in \mathbb{R}} |J_f(\tau)| < \infty, \qquad \lim_{L \to \infty} \sup_{\tau \in \mathbb{R}} \frac{|R(\tau, L)|}{L} = 0.$$

Theorem 4.2 (Vanishing excess under the exact window law)**.** *If f satisfies the exact window law, then*

$$E_{\text{ex}}(f) = 0.$$

Proof. Divide the exact window law by L:

$$\frac{Q_{\text{bank}}(f; \tau, L) - E_0 L}{L} = \frac{J_f(\tau + L) - J_f(\tau)}{L} + \frac{R(\tau, L)}{L}.$$

Take absolute values and then the supremum over τ:

$$\sup_{\tau \in \mathbb{R}} \left| \frac{Q_{\text{bank}}(f; \tau, L) - E_0 L}{L} \right| \leq \sup_{\tau \in \mathbb{R}} \left| \frac{J_f(\tau + L) - J_f(\tau)}{L} \right| + \sup_{\tau \in \mathbb{R}} \frac{|R(\tau, L)|}{L}.$$

Since J_f is bounded,

$$\sup_{\tau \in \mathbb{R}} \left| \frac{J_f(\tau + L) - J_f(\tau)}{L} \right| \leq \frac{2 \sup_\tau |J_f(\tau)|}{L} \to 0.$$

By assumption,

$$\sup_{\tau \in \mathbb{R}} \frac{|R(\tau, L)|}{L} \to 0.$$

Therefore

$$\limsup_{L \to \infty} \sup_{\tau \in \mathbb{R}} \frac{Q_{\text{bank}}(f; \tau, L) - E_0 L}{L} = 0.$$

Hence

$$E_{\text{ex}}(f) = 0.$$

□

5 Off-critical drift and quartic penalty

Definition 5.1 (Lifted drift)**.** Let

$$\rho = \beta + i\gamma$$

be a nontrivial zero of ζ. In log-time $\tau = \log t$, define the lifted drift

$$\delta_\rho(\tau) := \frac{\beta - \frac{1}{2}}{2\pi} \tau + r_\rho(\tau),$$

where r_ρ is bounded.

Definition 5.2 (Quartic drift penalty)**.** Define

$$P_4(\rho;\tau,L) := \int_\tau^{\tau+L} |\delta_\rho(u)|^4\,du.$$

Lemma 5.3 (Coercive quartic growth)**.** *Assume $\beta \neq \frac{1}{2}$ and r_ρ is bounded. Then there exist constants $c, C > 0$ such that for all sufficiently large τ and all $L \geq 1$,*

$$P_4(\rho;\tau,L) \geq c\left|\beta - \frac{1}{2}\right|^4 \tau^4 L - C\tau^3 L.$$

Proof. Write

$$\delta_\rho(u) = au + b(u), \qquad a := \frac{\beta - \frac{1}{2}}{2\pi} \neq 0,$$

with $b(u) := r_\rho(u)$ bounded. There exists $M > 0$ such that

$$|b(u)| \leq M \qquad \forall u.$$

Hence

$$|\delta_\rho(u)| = |au + b(u)| \geq |a|u - M.$$

For sufficiently large u, the right-hand side is positive, and

$$|\delta_\rho(u)|^4 \geq (|a|u - M)^4.$$

Expanding the quartic polynomial gives

$$(|a|u - M)^4 = |a|^4u^4 - 4|a|^3Mu^3 + 6|a|^2M^2u^2 - 4|a|M^3u + M^4.$$

Thus, for sufficiently large u,

$$|\delta_\rho(u)|^4 \geq c_1|a|^4u^4 - C_1u^3$$

for suitable constants $c_1, C_1 > 0$. Integrating over $[\tau, \tau + L]$,

$$P_4(\rho;\tau,L) \geq c_1|a|^4 \int_\tau^{\tau+L} u^4\,du - C_1 \int_\tau^{\tau+L} u^3\,du.$$

For $u \in [\tau, \tau + L]$ we have $u \geq \tau$, so

$$\int_\tau^{\tau+L} u^4\,du \geq \tau^4 L.$$

Also, for $L \geq 1$ and large τ,

$$\int_\tau^{\tau+L} u^3\,du \leq C_2\tau^3 L$$

for some constant $C_2 > 0$. Hence

$$P_4(\rho;\tau,L) \geq c|a|^4\tau^4 L - C\tau^3 L.$$

Since $|a| = \frac{1}{2\pi}\left|\beta - \frac{1}{2}\right|$, this gives

$$P_4(\rho;\tau,L) \geq c\left|\beta - \frac{1}{2}\right|^4 \tau^4 L - C\tau^3 L.$$

□

6 Bridge inequality and positive excess

Proposition 6.1 (Target bridge inequality)**.** *Let $\rho = \beta + i\gamma$ be an off-critical zero with $\beta \neq \frac{1}{2}$. A sufficient bridge for positive excess is the existence of constants $c_1 > 0$, $c_2 \geq 0$ such that*

$$Q_{\text{bank}}(f; \tau, L) - E_0 L \geq c_1 P_4(\rho; \tau, L) - c_2 L$$

for all sufficiently large τ and all $L \geq 1$.

Theorem 6.2 (Conditional positive excess)**.** *If the target bridge inequality holds for some off-critical zero ρ, then*

$$E_{\text{ex}}(f) > 0.$$

Proof. Divide the bridge inequality by L:

$$\frac{Q_{\text{bank}}(f; \tau, L) - E_0 L}{L} \geq c_1 \frac{P_4(\rho; \tau, L)}{L} - c_2.$$

By the coercive quartic growth lemma,

$$\frac{P_4(\rho; \tau, L)}{L} \geq c \left|\beta - \frac{1}{2}\right|^4 \tau^4 - C\tau^3$$

for sufficiently large τ. Hence

$$\frac{Q_{\text{bank}}(f; \tau, L) - E_0 L}{L} \geq c_1 \Big(c \left|\beta - \frac{1}{2}\right|^4 \tau^4 - C\tau^3 \Big) - c_2.$$

The right-hand side tends to $+\infty$ as $\tau \to \infty$. Therefore for sufficiently large τ, it is strictly positive. Taking $\sup_{\tau \in \mathbb{R}}$ and then $\limsup_{L \to \infty}$, we obtain

$$E_{\text{ex}}(f) > 0.$$

□

Remark 6.3. A stronger version yields $E_{\text{ex}}(f) = +\infty$ if one uses the full polynomial lower bound coming from

$$\int_{\tau}^{\tau+L} u^4 \, du = \tau^4 L + 2\tau^3 L^2 + 2\tau^2 L^3 + \tau L^4 + \frac{1}{5} L^5.$$

For the contradiction principle, however, strict positivity already suffices.

7 Conditional RH theorem

Theorem 7.1 (RH from closure plus bridge)**.** *Assume:*

(i) *the exact window law holds for f, hence $E_{\text{ex}}(f) = 0$;*

(ii) *the target bridge inequality holds for every off-critical zero.*

Then every nontrivial zero of $\zeta(s)$ satisfies

$$\Re(s) = \frac{1}{2}.$$

Proof. By the vanishing excess theorem,

$$E_{\mathrm{ex}}(f) = 0.$$

Assume for contradiction that there exists an off-critical zero

$$\rho = \beta + i\gamma, \qquad \beta \neq \frac{1}{2}.$$

Then by the conditional positive excess theorem,

$$E_{\mathrm{ex}}(f) > 0.$$

This contradicts $E_{\mathrm{ex}}(f) = 0$. Therefore no off-critical zero exists. Hence every nontrivial zero lies on the critical line:

$$\Re(\rho) = \frac{1}{2}.$$

□

8 What is established and what is still missing

8.1 Established spine

The proof spine currently established is:

1. Gaussian self-duality:
$$e^{-\pi x^2} \longmapsto e^{-\pi \xi^2}.$$

2. Theta transformation:
$$\theta(t) = t^{-1/2}\theta(1/t).$$

3. Functional equation:
$$\xi(s) = \xi(1-s).$$

4. Exact window law:
$$E_{\mathrm{ex}}(f) = 0.$$

5. Off-critical drift with quartic coercive growth:
$$\beta \neq \frac{1}{2} \Longrightarrow P_4(\rho; \tau, L) \text{ grows coercively.}$$

6. Target bridge inequality:
$$\beta \neq \frac{1}{2} \Longrightarrow E_{\mathrm{ex}}(f) > 0.$$

7. Contradiction principle:
$$E_{\mathrm{ex}}(f) = 0 \quad \text{and} \quad E_{\mathrm{ex}}(f) > 0 \quad \Rightarrow \quad \text{impossible.}$$

8.2 Exact missing tasks

To upgrade the conditional theorem to a fully unconditional theorem, the following items remain to be proved without hidden assumptions or circularity:

M1. Explicit spectral extraction. Write the off-critical contribution f_ρ in a form compatible with the chosen residue channel and dipole regularization.

M2. Transport formula. Derive the lifted drift law

$$\delta_\rho(\tau) = \frac{\beta - \frac{1}{2}}{2\pi}\tau + r_\rho(\tau)$$

with bounded remainder from the explicit off-critical component.

M3. Quantitative witness lower bound. Show that the finite witness bank detects the off-critical mode with a lower bound proportional to a coercive functional of the lifted drift.

M4. No-cancellation / no-hiding estimate. Show that the remainder channel cannot cancel the coercive contribution at the level required for the target bridge inequality.

M5. Uniformity in (τ, L). Verify that the estimates hold uniformly in the regime required by the definition of $E_{\mathrm{ex}}(f)$.

9 Final verdict

Theorem 9.1 (Current exact logical status)**.** *Within the present framework, the Riemann Hypothesis is reduced to the unconditional proof of the transport-to-bank bridge, i.e. to the unconditional verification of the missing tasks **M1–M5**.*

Equivalently, the current framework yields a rigorous conditional theorem:

$$\textit{Exact window law} \quad + \quad \textit{Target bridge inequality for every off-critical zero} \quad \Longrightarrow \quad \textit{RH.}$$

Proof. Everything before this point has already been proved in the previous sections. The only nontrivial step not yet promoted to unconditional status is the bridge from an off-critical spectral component to the required bank-level lower bound. Hence the entire remaining burden is precisely the set of missing tasks **M1–M5**. □

Remark 9.2. Compressed into one line:

$$\boxed{\text{closure is in place, obstruction is in place, the missing step is the transport-to-bank bridge.}}$$

Remark 9.3. Even shorter:

$$\boxed{\text{off-critical zero} \Longrightarrow \text{linear drift} \Longrightarrow \text{quartic penalty} \Longrightarrow E_{\mathrm{ex}}(f) > 0,}$$

and the only open issue is proving this chain unconditionally from the explicit residue channel.